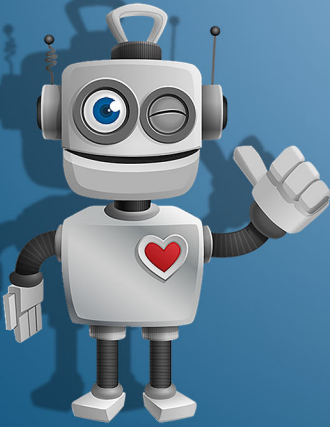


تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

اسئلة سنوات لمادة

نظرية التدم



ميد



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1	2	3	4	5	6	7	8	9	10	11	12	13	14
B	A	D	D	C	A	B	A	A	B	A	C	D	D

1. Let $F(s) = \frac{1}{s(s+1)}$, The final value of $f(t)$ is:

- a) 0 b) ∞ c) 1 d) None

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} \frac{1}{s+1} = 1$$

2. $y(t) = e^{-t} - e^{+t}$, Then $Y(s)$ will be

- A. $\frac{1}{s^2 + 1}$ B. $\frac{-2}{s^2 - 1}$ C. $\frac{+2}{s^2 - 1}$ D. None

3. Open-loop control can be used, in practice, only if:

- A. the relationship between the input and output is known B. there are no internal disturbances.
C. there are no external disturbances D. all above

4. The transfer function of an LC series circuit, where $v_o(t)$ is the output and the applied voltage $v(t)$ is the input, is

- A. $\frac{L}{LCs + 1}$ B. $\frac{1}{LCs^2 + 1}$ C. $\frac{Cs}{LCs^2 + 1}$ D. $\frac{1}{LCs + 1}$

5. The number of state equations in spring-mass-dashpot linear mechanical system is

- A. 4 B. 3 C. 2 D. 1

6. The output of the main comparator in a feedback control system is called

- A. error signal B. Feedback signal C. output signal D. Difference signal



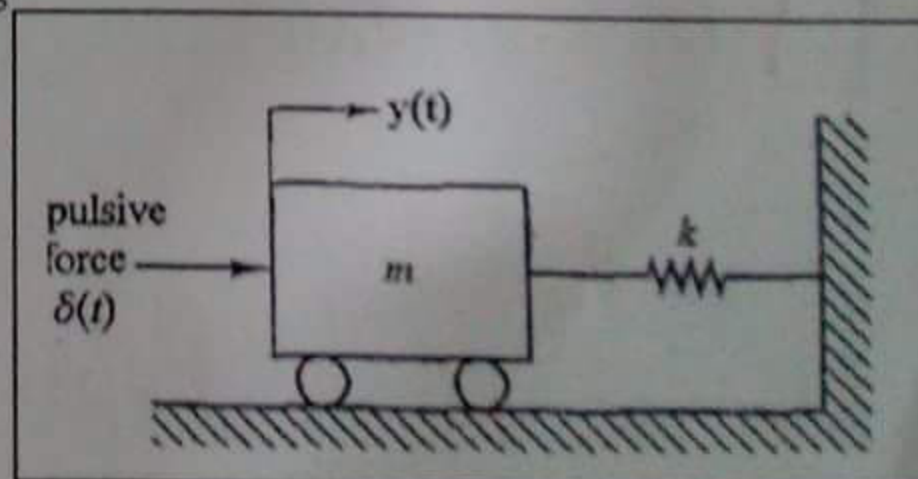
7. The major disadvantage of a feedback system may be

- A. Inaccuracy B. instability C. Unreliability D. inefficiency

8. For the system shown, the mathematical model relating

The output ($y(t)$) and the input ($\delta(t)$) is

- A. $my''(t) + fy'(t) + Ky(t) = \delta(t)$
B. $my''(t) + Ky(t) = \delta(t)$
C. $my''(t) + fy'(t) + Ky(t) = 0$
D. $my''(t) + Ky(t) = 0$



9. The state variables of a system given by $y'''(t) + 5y''(t) + 4y'(t) + 2y(t) = 3u(t)$ are

- A. $x_1(t) = y(t); x_2(t) = y'(t); x_3(t) = y''(t)$ B. $x_1(t) = 2y(t); x_2(t) = 4y'(t); x_3(t) = 5y''(t)$
C. $x_1(t) = y(t); x_2(t) = r(t)$ D. $x_1(t) = y(t); x_2(t) = y''(t)$

10. The steady state error in the unit ramp response of first order standard system is

- A. 0 B. T C. 2T D. 3T

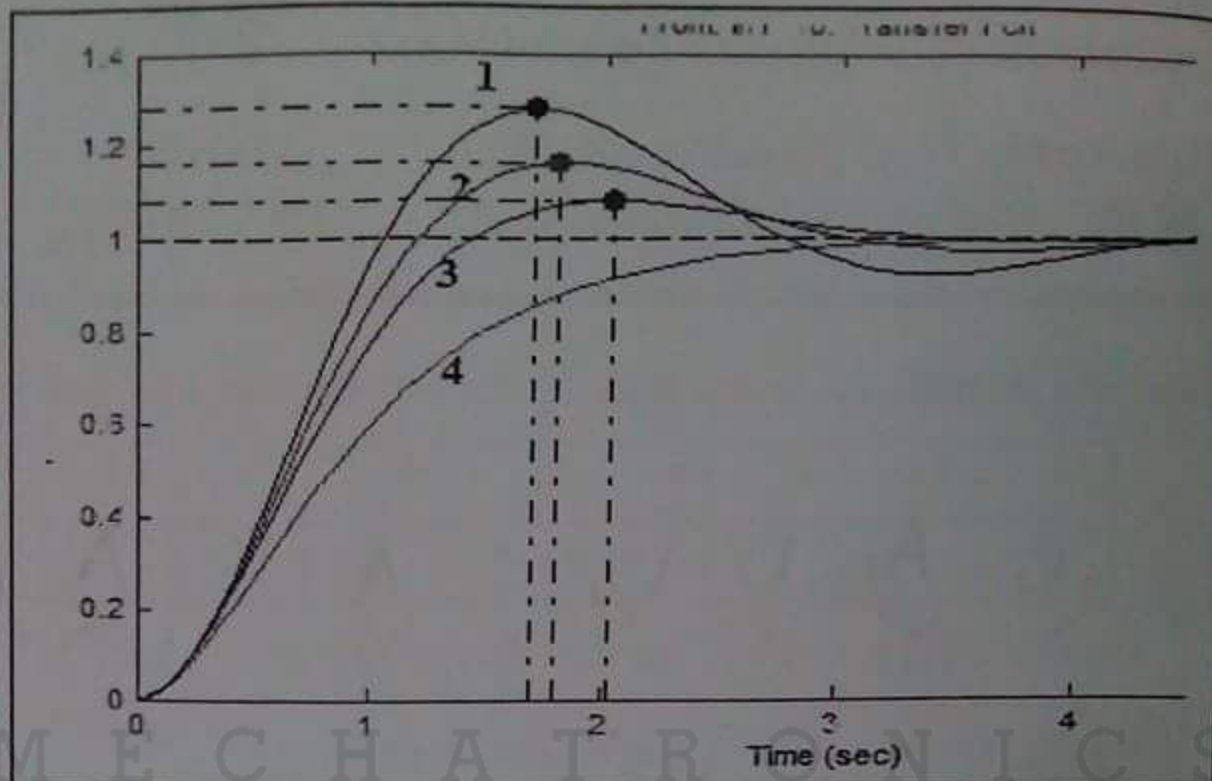
Depend on the following figure to solve 11 & 12.

11. The time response of a second Order system given by

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

is denoted as

- ☒ A. 1 B. 2 ☐ C. 3 D. 4



12. The unacceptable response with respect to the rise time is denoted by

- ☒ A. 1 B. 2 C. 3 D. 4

13. The partial fraction of $y(s) = \frac{1}{s^2(s+1)^2}$ is

A. $y(s) = \frac{1}{s^2(s+1)^2}$ B. 2 $y(s) = \frac{a}{(s+1)^2} + \frac{b}{s^2}$

☒ C. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{b}{s^2}$ ☐ D. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{\beta_1 s + \beta_2}{s^2}$ ← ans D

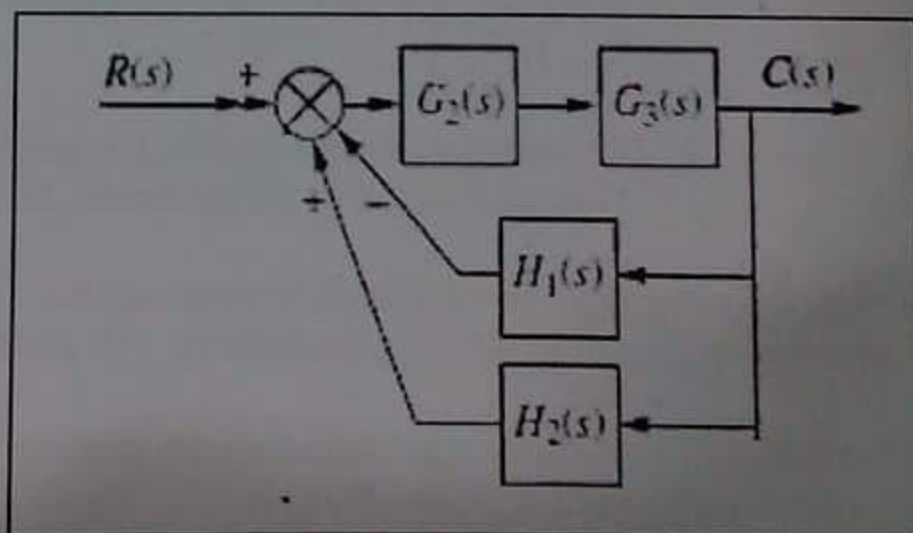
14. The open loop transfer function from the given block diagram is

A. $G_2 G_3 H_1 H_2(s)$

B. $G_2 G_3 H_1(s)$

C. $G_2 G_3 H_2(s)$

☒ D. $G_2 G_3 (H_2(s) - H_1(s))$



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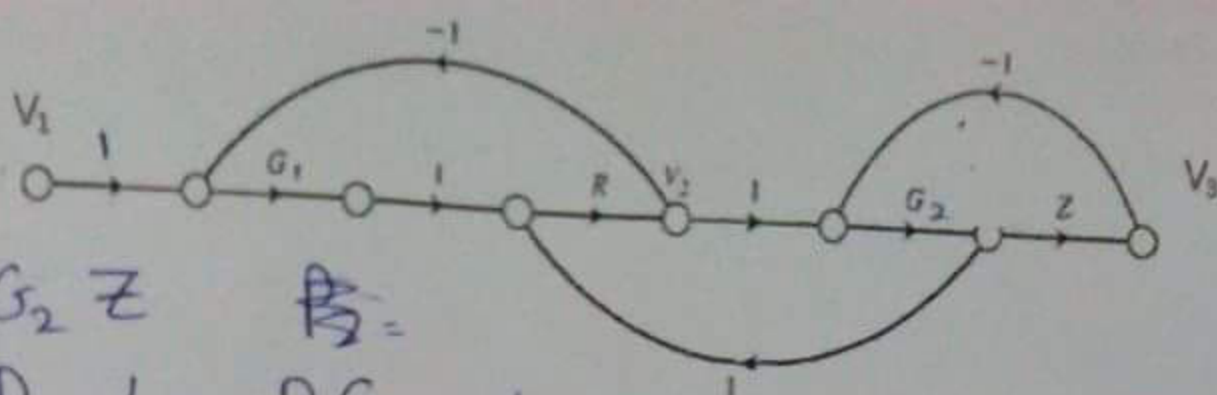


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$$P_1 = G_1 R G_2 Z$$

$$L_1 = -G_1 R \quad L_2 = R G_2 \quad L_3 = -G_2 Z$$

untouch loop

$$L_1 L_2 = -G_1 G_2 R^2$$

$$L_1 L_3 = +G_1 G_2 R Z$$

$$L_2 L_3 = -R G_2^2 Z$$

$$\Delta = 1 - [-G_1 R + R G_2 - G_2 Z] + [-G_1 G_2 R^2 + G_1 G_2 R Z - R G_2^2 Z]$$

$$\Delta_1 = 1$$

$$\frac{V_3}{V_1} = \frac{\Delta_1 \times P_1}{\Delta} = \frac{1 \times G_1 R G_2 Z}{1 - [G_1 R + G_2 R - G_2 Z] + [-G_1 G_2 R^2 + G_1 G_2 R Z - R G_2^2 Z]}$$

Q3) (6 marks) Assume zero initial conditions, solve the following differential equation using Laplace transform

$$y''(t) + 3y'(t) + 2y(t) = 1$$

$$s^2 y(s) + 3s y(s) + 2y(s) = \frac{1}{s}$$

$$y(s) [s^2 + 3s + 2] = \frac{1}{s}$$

$$y(s) = \frac{1}{s[s^2 + 3s + 2]} \Rightarrow \frac{1}{s(s+2)(s+1)} \Rightarrow \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$$

$$A(s+2)(s+1) + B(s+1)s + C(s+2)s = 1$$

$$\int \frac{1/2}{s} + \int \frac{1/2}{s+2} + \int \frac{-1}{s+1}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{2} e^{-2t} - e^{-t}$$